

Mathy bits

Formal descriptions of the operations that are in the code, explicitly or implicitly, with paste-ready MathJax. Audience: me. Each entry is *what it is* plus *the markup*. Source pointers are module-qualified, not file paths.

Claims marked (**checked**) were verified exhaustively over the relevant finite domain, or over the symbol table, at the time of writing.

0. Notation used throughout

symbol	meaning
T, I	tone set, interval set of a tonal system
V, V^o	WSMN abstract and octave-qualified tonal vectors
$\chi(d)$	chromatic value of the natural letter d (<code>DIATONES[d].chromatic</code>)
α	accidental alteration, half-steps from natural
ι	semitone map (<code>__int__</code>)
φ	line-of-fifths map (<code>fifths_position</code>)
$\mathcal{D}_S$	durations of temporal system S
ℓ	rational length (<code>rational_length</code>)
G, G_t	the score graph, its timing subgraph
s	the NEXT successor partial function

| `\chi(d)` `\quad` `\alpha` `\quad` `\iota` `\quad` `\varphi` `\quad` `\ell` `\quad` `\mathcal{D}_S` `\quad` `G_t`

1. The abstract contract

1.1 Tonal system

A tone set, an interval set, a transposition action, a distance. `Tone`, `Interval`, `TonalSystem`.

$$\text{transpose} : T \times I \rightarrow T, \quad \text{distance} : T \times T \rightarrow I$$

The directed version is what round-trips:

$$\text{transpose}(a, \text{diff}(b, a)) = b$$

In OMK, `distance` is the *undirected* magnitude, so this law holds for `b - a`, **not** for `a.distance(b)`. See §2.5.

1.2 Temporal system

A set of notated lengths with a rational measure. `TemporalSystem`, `TemporalElement`, `Measurable`.

$$\ell : \mathcal{D}_S \rightarrow \mathbb{Q}$$

1.3 System compatibility

The guard that stops two systems being mixed silently. `TonalSystem.compatible_with`, `TemporalSystem.compatible_with`.

$$S \frown S' \iff S = S' \vee \text{univ}(S) \vee \text{univ}(S')$$

$\frown$ is reflexive and symmetric but **not** transitive: the universal system (`ANY_TONAL_SYSTEM`, `ZeroDuration`) is compatible with everything, which is exactly what breaks transitivity.

| `S \frown S' \iff S = S' \ \lor\ \mathrm{univ}(S) \ \lor\ \mathrm{univ}(S')`

2. WSMN tonal algebra

2.1 The carrier

A tonal vector is a diatonic and a chromatic coordinate, optionally with an octave. `TonalVector`, `tonal_arithmetic`.

$$V = \mathbb{Z}_7 \times \mathbb{Z}_{12}, \quad V^o = \mathbb{Z}_7 \times \mathbb{Z}_{12} \times \mathbb{Z}$$

Octave carry comes from the diatonic coordinate alone, so the octave-qualified vectors are just $\mathbb{Z} \times \mathbb{Z}_{12}$ in disguise (**checked**):

$$\delta : V^o \xrightarrow{\sim} \mathbb{Z} \times \mathbb{Z}_{12}, \quad (d, c, o) \mapsto (d + 7o, c)$$

$\delta_1(x) = d + 7o$ is the staff position; `_diatonic_position`.

2.2 Addition

`tonal_sum`, `TonalVector.__add__`. Componentwise, with the diatonic overflow pushed into the octave.

$$(d_1, c_1) + (d_2, c_2) = (d_1 + d_2 \bmod 7, c_1 + c_2 \bmod 12)$$

$$(d_1, c_1, o_1) + (d_2, c_2) = \left(\overline{d_1 + d_2}, \overline{c_1 + c_2}, o_1 + \left\lfloor \frac{d_1 + d_2}{7} \right\rfloor \right)$$

Under δ this is plain componentwise addition in $\mathbb{Z} \times \mathbb{Z}_{12}$ (**checked**).

V is an abelian group; V^o is a V -set under $+$, and the projection is equivariant. Arities do not mix freely: a qualified vector may be moved by an abstract interval, but not the other way round (**TypeError**: an octave cannot be added to an abstract value).

$$+ : V^o \times V \rightarrow V^o, \quad \pi(x + i) = \pi(x) + i$$

| `(d_1, c_1, o_1) + (d_2, c_2) = \Bigl(\, \overline{d_1 + d_2}, \, \overline{c_1 + c_2}, \, o_1 + \, \Bigl\lfloor \frac{d_1 + d_2}{7} \Bigr\rfloor \, \Bigr)`

2.3 Alteration and the semitone map

`_tonal_unmodulo`, `tonal_int`, `_PitchRepresentation.alteration`.

The chromatic coordinate is read as the alteration *nearest the letter's natural*, i.e. lifted into a window of ± 6 :

$$\tilde{c}(d, c) = \text{the representative of } c \bmod 12 \text{ in } [\chi(d) - 6, \chi(d) + 6]$$

$$\alpha(d, c) = \tilde{c}(d, c) - \chi(d), \quad \iota(d, c, o) = \tilde{c}(d, c) + 12o$$

So one vector carries up to six half-steps of alteration either way ($D\sharp$ against $F\flat$ is a quadruply diminished third).

Homomorphism, conditionally (checked):

$$\iota(x + y) \equiv \iota(x) + \iota(y) \pmod{12} \quad \text{for all abstract } x, y$$

$$\iota(x + y) = \iota(x) + \iota(y) \quad \text{for qualified } x, y \text{ with } |\alpha| \leq 3$$

It fails only where the ± 6 window wraps — i.e. where the spelling runs off the end of the accidental table. Worth saying in the paper: the semitone map is a homomorphism *on the musically inhabited part of the carrier*, and the failure is a fact about notation, not about pitch.

2.4 Enharmonic equivalence as a quotient

Code equality is equality in the carrier: spelling-sensitive. Enharmonic equivalence is the strictly coarser quotient by ι .

$$x \sim_{\text{enh}} y \iff \iota(x) = \iota(y), \quad V^o / \sim_{\text{enh}} \cong \mathbb{Z}, \quad V / \sim_{\text{enh}} \cong \mathbb{Z}_{12}$$

$$x = y \implies x \sim_{\text{enh}} y, \quad x \sim_{\text{enh}} y \not\implies x = y$$

$C\sharp$ and $D\flat$ are enharmonic and unequal; `hash` agrees with `=`, not with `~`.

2.5 Transposition, difference, distance

`transpose`, `__add__`, `__sub__`, `distance` (`tonal_abs_diff`).

$$\text{transpose}(a, i, \uparrow) = a + i, \quad \text{transpose}(a, i, \downarrow) = a - i$$

The directed difference is the one with the torsor law (**checked**, all 7056 abstract pairs):

$$a + (b - a) = b$$

`distance` is the undirected magnitude — the smaller of the two directed differences, spelled as an interval:

$$\text{dist}(a, b) = \min_{\leq} \{ \text{ic}(b - a), \text{ic}(a - b) \}, \quad \text{dist}(a, b) = \text{dist}(b, a)$$

(**checked**) symmetric everywhere; on abstract vectors with $|\alpha| \leq 1$ it is bounded by the tritone:

$$|\iota(\text{dist}(a, b))| \leq 6$$

And the caveat worth a sentence in the paper: $a + \text{dist}(a, b) = b$ fails in general (it holds only when b is above a and there is no enharmonic tie), because the magnitude has thrown away the direction and the spelling. $C4$ and $B\sharp3$ are a distance of a diminished second apart in either direction, and neither one is the other transposed by it.

2.6 Inversion

`tonal_invert`, `TonalVector.inversion`. Reflection through a centre.

$$\text{inv}_y(x) = y - (x - y) = 2y - x$$

(**checked**) an involution, and equal to the affine reflection, over all 7056 pairs:

$$\text{inv}_y \circ \text{inv}_y = \text{id}, \quad \text{inv}_0(x) = -x$$

Interval inversion in the classical sense is inversion about the origin, which in V is just negation: a major third $(2, 4)$ inverts to a minor sixth $(5, 8) = -(2, 4)$.

$$\begin{array}{|l} \backslash\operatorname{operatorname{inv}}\}_{\{y\}}(x) = y - (x - y) = 2y - x, \quad \backslash\operatorname{qqquad} \\ \hookrightarrow \backslash\operatorname{operatorname{inv}}\}_{\{y\}}\backslash\operatorname{circ}\backslash\operatorname{operatorname{inv}}\}_{\{y\}} = \mathrm{id} \end{array}$$

2.7 Interval class

`abs_interval`: the smaller of an interval and its inversion.

$$\mathrm{ic}(x) = \min_{\preceq} \{ x, -x \}$$

2.8 Order

`--lt--`, `--gt--` compare by semitone only, so they are a **preorder** whose induced equivalence is exactly enharmonic equivalence:

$$x \prec y \iff \iota(x) < \iota(y)$$

`tonal_higher_of` / `tonal_lower_of` break the enharmonic tie by staff position, which refines it to a **total order (checked, injective on both V and V^o)**:

$$x \sqsubset y \iff (\iota(x), \delta_1(x)) <_{\mathrm{lex}} (\iota(y), \delta_1(y))$$

`tonal_larger_of` / `tonal_smaller_of` are the same thing on magnitudes (interval size rather than pitch height).

2.9 Octave qualification

`qualify_octave`, `unqualify_octave`, `conditional_qualify_octave`: a projection and a family of sections.

$$\pi : V^o \rightarrow V, (d, c, o) \mapsto (d, c); \quad \sigma_o : V \rightarrow V^o, (d, c) \mapsto (d, c, o); \quad \pi \circ \sigma_o = \mathrm{id}_V$$

2.10 Nearest instance

`tonal_nearest_instance`: pick the representative of b 's fibre closest to a , ties to the smaller diatonic step (Lilypond's relative-octave rule).

$$\mathrm{near}(a, b) = \arg \min_{b' \in \pi^{-1}(\pi(b))} (|\iota(a) - \iota(b')|, |\delta_1(a) - \delta_1(b')|)$$

$$\pi(\mathrm{near}(a, b)) = \pi(b), \quad |\iota(a) - \iota(\mathrm{near}(a, b))| \leq 6$$

2.11 The line of fifths

`fifths_position`. Naturals step by 2 (mod 7, re-centred on C), each sharp adds 7. The line does **not** wrap — $B_\sharp$ is 12, not 0 — because spelling is what is being counted.

$$\varphi(d, c) = ((2d + 1) \bmod 7) - 1 + 7\alpha(d, c)$$

A group homomorphism (checked, exact for all $|\alpha| \leq 3$, which is the whole representable key-signature range):

$$\varphi(x + y) = \varphi(x) + \varphi(y), \quad \varphi : V \rightarrow \mathbb{Z}$$

Read as a pitch, φ is the signed number of sharps in the major key on that tonic. Read as an interval, it is how far a key signature travels around the circle when its tonic moves by that interval. This is the cleanest statement in the tonal half: *transposition acts on key signatures by translation on the line of fifths.*

$$\begin{aligned} \varphi(d, c) &= \text{bigl}((2d+1) \bmod 7 \text{bigr}) - 1 + 7 \backslash, \alpha(d, c), \quad \varphi(x+y) = \\ &\hookrightarrow \varphi(x) + \varphi(y) \end{aligned}$$

3. Lifting a tonal operation

3.1 Pointwise action

Every `transform` / `transform_tones` in the codebase is the same thing: a tone operation pushed through a structure pointwise. `apply_tone_operation` is the single guard (`TypeError` unless the result is a `Tone`).

$$f_*(t_1, \dots, t_n) = (f(t_1), \dots, f(t_n)), \quad (g \circ f)_* = g_* \circ f_*, \quad (\text{id})_* = \text{id}$$

Carriers: `ToneCollection.transform`, `NoteEvent.transform_tones`, `Chord.transform`, `KeySignature.transform`, `Key.transform`, `ModalContextEvent.transform_tones`, `OmkGraph.transform_tones`.

Universal-system tones are fixed points (a rest stays a rest):

$$f(t) = t \quad \text{whenever } \text{univ}(\text{sys}(t))$$

3.2 Tone collections

`ToneCollection`: an *ordered* tuple with an optional root and a name template. Order is musically significant (added 2nd vs. added 9th), so this is a sequence, not a set.

$$\mathbf{t} = (t_1, \dots, t_n), \quad r \in T \cup \{\perp\}$$

$$f_*(\mathbf{t}, r) = (f_*\mathbf{t}, f(r))$$

Equality ignores the name:

$$(\mathbf{t}, r, \nu) = (\mathbf{t}', r', \nu') \iff \mathbf{t} = \mathbf{t}' \wedge r = r'$$

Subsets: `combinations(k)` and `all_combinations` are order-preserving k -subsets.

$$\binom{\mathbf{t}}{k}, \quad \left| \bigcup_{k=2}^{n-1} \binom{\mathbf{t}}{k} \right| = 2^n - n - 2$$

3.3 Chords

`ChordType` is an interval pattern rooted at the origin; `Chord` is that pattern realized at a root. Realization is the group action.

$$\tau \subseteq V, \quad 0 \in \tau, \quad b \in \tau \quad \tau(r) = \{r + t : t \in \tau\}, \quad \text{bass } b + r$$

Transposition commutes with realization — which is why `ChordType.transform` raises rather than doing anything:

$$T_i(\tau(r)) = \tau(r + i)$$

Inversion is a change of bass, not a change of tones; arpeggiation is the cyclic rotation that brings the bass to the front.

$$\rho_b(t_1, \dots, t_n) = (t_k, \dots, t_n, t_1, \dots, t_{k-1}), \quad t_k = b$$

Slash notation C/E is `__truediv__`, i.e. $\tau \mapsto (\tau, b')$.

3.4 Key signatures

KeySignature: an alteration per letter.

$$\kappa : \mathbb{Z}_7 \rightarrow \{-3, \dots, 3\}$$

fifths is defined only on *standard* signatures — those whose values, read in sharp order $F C G D A E B$, take at most two distinct values differing by one and are non-increasing:

$$\text{fifths}(\kappa) = \sum_{d \in \mathbb{Z}_7} \kappa(d) \quad (\kappa \text{ standard})$$

from_alts is its section: $\text{fifths}(\kappa_n) = n$.

Transformation runs each letter through f as a pitch and reads the alteration back off the image letter; it is defined only when the induced letter map is a bijection of $\mathbb{Z}_7$ (otherwise two letters collide and there is no signature):

$$\kappa'(\pi_1 f(d, \chi(d) + \kappa(d))) = \alpha(f(d, \chi(d) + \kappa(d)))$$

3.5 Keys and modes

ModePattern is an interval pattern through the origin; a **Key** is a mode at a tonic, plus a signature.

$$M \subseteq V, \ 0 \in M \quad K = (r, M, \kappa), \quad \text{tones}(K) = M(r)$$

Key.transform keeps the mode:

$$f \cdot (r, M, \kappa) = (f(r), M, f_* \kappa)$$

Worth a remark in the paper: for translations this agrees with transforming the tones, and for everything else it does not. C major inverted about C is C major by “keep the mode” and C Phrygian by “transform the tones”. The code picks the first; the divergence is exactly the statement that a mode is a pattern, not a set.

A bare signature (no tonic) moves with the music; **NoKey** (no tonic, no signature) is a fixed point of every f .

4. Temporal algebra

4.1 Length

Measurable.rational_length. Additive, sign-preserving, surjective onto the positives.

$$\ell : \mathcal{D}_S \rightarrow \mathbb{Q}, \quad \ell(a + b) = \ell(a) + \ell(b), \quad \ell(-a) = -\ell(a), \quad \ell(\mathbf{0}) = 0$$

(checked over the symbol table) addition is commutative and associative, $a + (-a) = \mathbf{0}$, and $\mathbf{0}$ is a two-sided identity across systems.

$(\mathcal{D}_S, +, \mathbf{0})$ is a commutative monoid with negation; ℓ is a monoid homomorphism to $(\mathbb{Q}, +, 0)$.

4.2 Equality as a quotient

Durations compare and hash by length, not spelling: a dotted quarter *is* three eighths *is* `TemporalUnit(3, eighth)`.

$$a \equiv b \iff \ell(a) = \ell(b), \quad a < b \iff \ell(a) < \ell(b), \quad \mathcal{D}_S / \equiv \cong \ell(\mathcal{D}_S) \subseteq \mathbb{Q}$$

Sequences are measured by their sum, so a **Measurable** compares against any iterable of them:

$$\ell(\langle a_1, \dots, a_n \rangle) = \sum_i \ell(a_i)$$

Only within compatible systems (§1.3): incompatible elements are never equal, and ordering them is an error rather than a **False**.

$$\neg(S \frown S') \implies a \neq b \text{ and } a < b \text{ undefined}$$

4.3 Scaling

`Measurable.scale`: a partial action of the positive rationals.

$$\cdot : \mathcal{D}_S \times \mathbb{Q}_{>0} \rightarrow \mathcal{D}_S, \quad \ell(a \cdot k) = k \cdot \ell(a), \quad (a \cdot k) \cdot k^{-1} = a$$

(**checked**) exact for every symbol and every positive rational scalar; **ScalingError** for $k \leq 0$ or where the system cannot spell the result. Powers of two preserve the spelling; other scalars may change the notation (quarter $\times 3$ is a dotted half, quarter $\times \frac{2}{3}$ is a triplet quarter, quarter $\times 5$ is a tie).

4.4 The notatable values

The set of lengths writable as **one symbol**: a power-of-two base with k dots.

$$N = \left\{ 2^j (2^{k+1} - 1) 2^{-k} \mid j \in \mathbb{Z}, k \in \mathbb{Z}_{\geq 0} \right\}$$

4.5 Notation as a section of length

`MetricalDuration.from_length`: every positive rational gets a canonical spelling.

$$\nu : \mathbb{Q}_{>0} \rightarrow \mathcal{D}_{\text{WSMN}}, \quad \ell \circ \nu = \text{id}_{\mathbb{Q}_{>0}}$$

(**checked**) ν is a section of ℓ , and a right inverse on single symbols: $\nu(\ell(a)) = a$ with the same nominal value, dots and ratio.

The three rules, as a case split:

$$\nu(x) = \begin{cases} \text{the symbol } x & x \in N \\ \text{tuple } k : 2^{\lfloor \log_2 k \rfloor} & k = \text{odd}(\text{den}(x)) > 1 \\ \text{greedy tie} & \text{otherwise} \end{cases}$$

Greedy tie split, largest single symbol first:

$$x = \sum_{i=1}^m n_i, \quad n_i = \max \left\{ n \in N \mid n \leq x - \sum_{j<i} n_j \right\}$$

ν is canonical, not context-aware: it ignores the meter and the beat position, which is the honest limit to state in the paper.

4.6 Ratios: tuplets, tempo, modulation

TemporalRatio. One object for three jobs, distinguished only by which systems the two sides live in.

$$R = (\text{nominal}, \text{contextual}), \quad m(R) = \frac{\ell(\text{contextual})}{\ell(\text{nominal})}$$

$$\ell_{\text{real}}(d) = \ell_{\text{nom}}(d) \cdot m(R)$$

Equality and hashing are by multiplier, so a sextuplet 6:4 is the same ratio as a triplet 3:2 on any base (**checked**):

$$R \equiv R' \iff m(R) = m(R')$$

The three readings:

$$\begin{aligned} \text{tuplet: } & S \rightarrow S \text{ within one line} \\ \text{metric modulation: } & S \rightarrow S \text{ across a boundary} \\ \text{tempo: } & S \rightarrow \text{clock time} \end{aligned}$$

Tempo is the ratio against real time; the multiplier comes out in microseconds per whole note.

$$\text{Tempo}(n, \beta) = (\langle n, \beta \rangle, \langle 1, 60 \text{ s} \rangle), \quad m = \frac{6 \times 10^7}{n \cdot \ell(\beta)}$$

Conversion is then $\mathbb{Q}$ -linear — a change of unit, nothing more:

$$\begin{aligned} \mathcal{C}_R : \mathcal{D}_S \rightarrow \mathcal{D}_{\text{clock}}, \quad \mathcal{C}_R(d) &= \ell(d) \cdot m(R) \\ \left| \begin{array}{l} m(R) = \frac{\ell(\text{contextual})}{\ell(\text{nominal})}, \quad \mathcal{C}_R(d) = \\ \hookrightarrow \ell(d) \cdot m(R) \end{array} \right. \end{aligned}$$

4.7 Meters

TemporalUnit, **CompoundTemporalUnit**, **TimeSignature**. A bar is an ordered series of counted units; additive meters are several.

$$U = (n, \beta), \quad \ell(U) = n \ell(\beta) \quad M = \langle U_1, \dots, U_k \rangle, \quad \ell(M) = \sum_i \ell(U_i)$$

$$\text{rem}(M, \langle s_i \rangle) = \ell(M) - \sum_i \ell(s_i)$$

$$\text{oob}(M, \langle s_i \rangle) = \min \left\{ i \mid \sum_{j \leq i} \ell(s_j) > \ell(M) \right\}$$

Time signatures compare by total length alone: $4/4 = 2/2 = 8/8$ (**checked**).

4.8 The zero and the graces

ZeroDuration is the identity of every system; **GraceDuration** is an element of $\ker \ell$ that still carries a notated symbol.

$$\ell(g) = 0, \quad a + g = g + a = a, \quad -g = g$$

So all graces are equal to each other and to zero; **nominal** is the extra datum that distinguishes them.

A run of graces adds nothing to the length of a line — which is precisely the statement that they live in the kernel.

5. Graph operations

5.1 The object

OmkgGraph over a **GraphAdapter**. A labelled multidigraph whose nodes are score objects.

$$G = (N, E, \text{src}, \text{tgt}, \tau), \quad \tau : E \rightarrow \Lambda$$

Λ is **EdgeType**; at most one edge of a given type between an ordered pair.

Node identity is a UUID; node *equality* is musical content, with **id** and **meta** excluded. So two separately built **B** quarter notes are equal and distinct:

$$u = v \not\Rightarrow u \equiv_{\text{id}} v$$

5.2 Lines

NEXT is a partial injective function: in-degree and out-degree at most one.

$$\text{deg}_{\text{NEXT}}^+(v) \leq 1, \quad \text{deg}_{\text{NEXT}}^-(v) \leq 1$$

so the **NEXT**-components are paths or cycles, and a *line* is a maximal chain. **walk_line** is the orbit of the successor s , truncated at the first repeat, which is what makes a cyclic line (a gamelan cycle) terminate:

$$\text{line}(v) = \langle v, s(v), s^2(v), \dots \rangle$$

5.3 Spans

walk_span: the line, plus everything branched off it, transitively. Pins are never crossed, so a span is the reachability closure under **NEXT** $\cup$ **BRANCHES**.

$$\text{span}(v) = \text{line}(v) \cup \bigcup_{u \in \text{line}(v)} \bigcup_{h \in B(u)} \text{span}(h)$$

The shared visited-set makes this idempotent and terminating even when two lines branch into each other.

$$\text{line}(v) \subseteq \text{span}(v)$$

5.4 The timing graph

This is the one I most want to get right in the paper.

Timing is a weight function on a subgraph, and a position is *always* relative — there is no origin.

$$w(u, v) = \begin{cases} \text{dur}(u) & v = s(u) \\ \text{disp}(e) & e : u \rightarrow v \text{ timed, anchor}(e) = \text{onset} \\ \text{dur}(u) + \text{disp}(e) & e : u \rightarrow v \text{ timed, anchor}(e) = \text{offset} \end{cases}$$

with $w(v, u) = -w(u, v)$, and the edge simply **absent** where **dur** is undefined: an event of unknown duration is opaque, and nothing after it or hanging from its offset has a known position.

$$w : E_t \rightarrow \mathcal{D}, \quad \text{dur}(u) = \perp \implies (u, s(u)) \notin E_t$$

Relative onset is a path sum (`_relative_onset`, breadth-first):

$$\text{on}(r, v) = \sum_{e \in P} w(e), \quad P : r \rightsquigarrow v$$

$$\text{on}(r, r) = \mathbf{0}, \quad \text{on}(u, v) = -\text{on}(v, u)$$

(**property-tested**) antisymmetric, zero on the diagonal, and equal to the sum of the intervening durations along a line.

5.5 Consistency as a coboundary

The timing graph is consistent exactly when w is a coboundary — when a global potential exists.

$$\exists \phi : N \rightarrow \mathcal{D} \text{ with } w(u, v) = \phi(v) - \phi(u) \iff \sum_{e \in C} w(e) = \mathbf{0} \text{ for every cycle } C$$

NEXT, BRANCHES and group CONTAINS edges form a **forest**, so a potential always exists and no check is needed. Pins (**Simultaneous**) are the only edges that can close a cycle, so they are the only source of inconsistency.

`check_alignment` measures the holonomy of each such cycle, edge by edge:

$$\text{conflict}(e) \iff w(e) \neq \text{on}_{G_t \setminus e}(\text{src}(e), \text{tgt}(e))$$

and reports rather than raises: a disagreement between a pin and the durations is a musical fact about an incomplete source, not a malformed graph.

$$\left| \begin{array}{l} \exists \phi : \forall u, v \quad w(u, v) = \phi(v) - \phi(u) \iff \sum_{e \in C} w(e) = \mathbf{0} \quad \forall C \end{array} \right|$$

5.6 Simultaneity, asserted and derived

Two notions, deliberately distinct.

Asserted — a pin with no displacement, symmetric in meaning, stored directed:

$$u \parallel v \iff \exists e : u \rightarrow v, \tau(e) = \text{SIMULTANEOUS}, \text{anchor}(e) = \text{onset}, \text{disp}(e) = \perp$$

Derived — equal onsets, computed:

$$u \equiv_t v \iff \text{on}(u, v) = \mathbf{0}$$

$\equiv_t$ is an equivalence relation on each connected component of G_t ; $\parallel$ is not transitive and not an equivalence — it is a *claim*, and $\parallel \subseteq \equiv_t$ is exactly the condition `check_alignment` verifies.

$$u \parallel v \implies u \equiv_t v \quad (\text{when consistent})$$

5.7 Containment, ownership, origin

Three different containments, distinguished by what they assert about timing:

BRANCHES : same performer; span-inclusive; timed
CONTAINS (LineGroup) : timing origin; not span-inclusive
CONTAINS (Score) : membership only; untimed

A **Score** is deliberately *not* a timing origin: two of its lines with no pin between them have no relative onset at all.

$$\text{on}(u, v) = \perp \quad \text{for } u, v \text{ in the same Score, unlinked in } G_t$$

5.8 Stints and materialization

A **Stint** is a closed interval of a line in span order, a **Part**'s run on it.

$$S = [a, b] \subseteq \text{span}(a), \quad b = \perp \Rightarrow \text{to the end of the line}$$

materialize forks a doubled line: it copies the **induced subgraph** on the covered content and re-points the stint at the copy.

$$H = G[\text{content}(S)], \quad H' \cong H, \quad G \rightsquigarrow G \sqcup H'$$

with the stint's **STARTS_AT/ENDS_AT** moved to H' . Content is the events plus their transitive attachments (marks, spanners, annotations, syllables); a spanner with an endpoint outside $\text{content}(S)$ is dropped, because the copy has no node for its other end. Not copied: relations to the rest of the graph. So the fork is exact and *local*:

$$\text{span}(a) \text{ in } G \text{ unchanged,} \quad f_* \text{ on } H' \text{ does not touch } H$$

5.9 Transformation over a span

`OmkGraph.transform_tones`: the pointwise action of §3.1, indexed by the span.

$$f \cdot G : \forall u \in \text{span}(a, b) \cap \text{TonalObject}, \quad u \mapsto f_*(u)$$

(**property-tested**) up then down is the identity, for every interval, over the line and its branches together; the pinned line is untouched, which is the graph-level statement that transposition follows ownership, not coincidence.

$$T_{-i} \cdot (T_i \cdot G) = G$$

5.10 Lyric binding

A **LYRIC** edge marks a syllable's **onset**, so the binding is a partial, order-preserving, injective map from events to syllables.

$$\lambda : E_{\text{line}} \rightarrow \Sigma$$

$$\text{dom } \lambda = \{ e \mid \ell(\text{dur}(e)) \neq 0, \neg \text{rest}(e), e \notin \text{interior}(\text{binding span}) \}$$

What is actually *sung* at an event is the step function that carries the last onset forward — which is what makes a melisma a fibre rather than a special case:

$$\hat{\lambda}(e) = \lambda \left(\max \{ e' \leq e \mid e' \in \text{dom } \lambda \} \right)$$

$$\text{melisma at } \sigma = \hat{\lambda}^{-1}(\sigma), \quad |\hat{\lambda}^{-1}(\sigma)| > 1$$

A reciting tone is the dual case: several syllables at one event, added explicitly with `connect_lyric_to_object`, so λ is injective only on the zipped construction, not by fiat.

6. Translation across systems

6.1 Temporal

A `TemporalRatio` whose two sides are in different systems is a change of unit. This is the *only* way durations cross a system boundary; without one, comparison returns `False` and ordering raises.

$$R_{S \rightarrow S'}, \quad \mathcal{C}_R : \mathcal{D}_S \rightarrow \mathcal{D}_{S'}, \quad \mathcal{C}_R(d) = \ell_S(d) \cdot m(R)$$

$$\mathcal{C}_{R'} \circ \mathcal{C}_R = \mathcal{C}_{R''}, \quad m(R'') = m(R) m(R')$$

One punctum of chant to one quarter note is the same construction as quarter = 120; only the codomain differs.

6.2 Tonal

Any $f : T_S \rightarrow T_{S'}$ may be pushed through the structures of §3 — `transform_tones` does not require the result to be of the same type, only to be a `Tone`. So translation between tonal systems is the same machinery as transposition, with a different f .

$$f : T_S \rightarrow T_{S'}, \quad f_* : \mathcal{C}(T_S) \rightarrow \mathcal{C}(T_{S'})$$

Universal tones are fixed (§3.1), so silence survives any translation.

6.3 Reading maps

`Stint.transposition` is not applied to the events — it is a map the consumer applies while reading. Two parts over one line read it differently without either owning it:

$$\text{read}_S(e) = T_{i_S}(\text{content}(e)), \quad i_S = \text{transposition}(S)$$

$$\text{read}_{S_1} \neq \text{read}_{S_2} \quad \text{over the same } e$$

A clarinet in A and a concert-pitch flute are two readings of one event, which is the point: the transposition lives on the relationship, not on the note.

7. Equality, collected

The paper will need this in one place: OMK uses at least seven distinct equality relations, each a quotient of something.

Tone / Interval : equality in the carrier (spelling-sensitive)

enharmonic : $\ker \iota$, strictly coarser

Duration : $\ker \ell$, spelling-*ins*sensitive, within $\curvearrowright$

TemporalRatio : $\ker m$

ToneCollection : $(\mathbf{t}, r)$, name excluded

Chord : $(\mathbf{t}, r, b)$, name and suffix excluded

OmkObject : musical content; id and meta excluded

OmkEdge : (τ, origin) ; id and meta excluded

The pattern worth naming: **values are equal when they mean the same thing, and the “same**

thing” is chosen per type. Durations quotient away notation; tones do not. That asymmetry is a claim about the two domains, not an inconsistency: a dotted quarter and three tied eighths sound identical, C_4 and D_4 do not behave identically.

Two invariants that go with it:

$$a = b \implies \text{hash}(a) = \text{hash}(b)$$

value types are hashable and frozen; graph objects are mutable and unhashable

8. Formatting scraps

Things I keep re-deriving.

Inline and display:

```
\(x + i\)
\[ \operatorname{transpose} : T \times I \to T \]
```

Named operators (avoids italic run-together):

```
\operatorname{transpose} \quad \operatorname{dist} \quad \operatorname{inv}_y \quad
\hookrightarrow \operatorname*{arg\,min}_{x \in X}
```

Maps and arrows:

```
f : A \to B \quad A \xrightarrow{f} B \quad A \hookrightarrow B \quad A
\hookrightarrow \twoheadrightarrow B
\quad A \rightharpoonup B \quad \text{(partial)} \quad A \xrightarrow{\sim} B
\hookrightarrow \quad \text{(iso)}
```

Modular and floor:

```
d_1 + d_2 \bmod 7 \quad \overline{d_1 + d_2} \quad \bigl\lfloor \tfrac{d}{7} \bigr\rfloor
\quad x \equiv y \pmod{12} \quad \mathbb{Z}_{12} \quad \mathbb{Z}/12\mathbb{Z}
```

Cases:

```
\[
w(u,v) =
\begin{cases}
\mathrm{dur}(u) \ \& \ v = s(u) \ \& \\
\mathrm{disp}(e) \ \& \ \mathrm{anchor}(e) = \text{onset} \ \& \\
\mathrm{dur}(u) + \mathrm{disp}(e) \ \& \ \mathrm{anchor}(e) = \text{offset}
\end{cases}
\]
```

Aligned multi-line:

```
\[
\begin{aligned}
\ell(a+b) \ \&= \ \ell(a) + \ell(b) \ \& \\
\ell(-a) \ \&= -\ell(a) \ \& \\
\ell(a \cdot k) \ \&= k \cdot \ell(a)
\end{aligned}
\]
```

Set-builder with sized delimiters:

```
\[
N = \Bigl\{ \, , \ 2^{\{ \, , j \} } \Bigr\} (2^{\{ \, , k+1 \} } - 1) \Bigr\} 2^{-k} \, ; \Bigm| \, ; \ j \in \mathbb{Z}, \ k \geq 0
\hookrightarrow \, , \Bigr\}
\]
```

A commuting square without tikz (MathJax-safe):

```
\[
\begin{array}{ccc}
```

```

V & \xrightarrow{\ T_i\ } & V \\
\downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi} \\
\mathbb{Z} & \xrightarrow{\ +\varphi(i)\ } & \mathbb{Z}
\end{array}
\]

```

Spacing, smallest to largest: `\`, `\:` `\;` `\quad` `\qquad`. Text inside math: `\text{...}`, or `\mathrm{...}` for short operator-ish names.

Not portable between MathJax and LaTeX

Things that compile in LaTeX but render as a red error in MathJax, so they cannot go in the markdown source:

```

\textsc{...}      % LaTeX only - use \mathrm{NEXT} for enum-ish names
\text{\emph{}}    % LaTeX only - use \textit{...} inside \text{...}

```

MathJax has no small caps at all, which is why the edge types are set as NEXT, BRANCHES, CONTAINS, SIMULTANEOUS — all-caps roman, matching how `EdgeType` spells them in the code.